

Outage Investigation Report —

OIR-2025-007

Field	Value
Document ID	OIR-2025-007
Report Date	2025-06-10
Author	Elena Vasquez, Reliability Engineer II
Reviewing Manager	Priya Ramachandran, Plant Manager, NW-PowerPool
Region	NW-PowerPool
Site	NW-PowerPool-WIND-003
Asset	TRA-10561 (Siemens Energy, transformer)
Outage ID	OUT-2025-1405
Classification	Forced outage — vibration_excess
Distribution	NW-PowerPool Operations, Reliability Engineering, Compliance, OEM Liaison

1. Outage Summary

TRA-10561 (NW-PowerPool-WIND-003 step-up transformer) tripped on tank-vibration relay action on 2025-05-25 at 21:07 PDT. Restoration completed 2025-06-01 at 11:07 PDT, total duration **158 hours**. Lost generation: **2,615.1 MWh**.

The event falls inside the Spring Vibration Cascade window (RCA-2025-001). Across May 2025 NW-PowerPool logged 19 forced outages, 11,360 MWh of lost generation. This event continues the pattern opened by OIR-2025-006 (WIN-10058) three weeks prior.

2. Timeline of Events

Timestamp (PDT)	Event
2025-05-24 04:30	Tank-vibration sensor V-12 records baseline shift from 2.1 mm/s to 3.0 mm/s. Per MS-2025-03 the "monitor" threshold is 3.5 mm/s.
2025-05-25 09:50	Sensor V-12 trends to 3.4 mm/s under 30 MVA load and 22 °C ambient.
2025-05-25 18:15	Top-oil temperature elevated 71 °C (normal 64–67 °C for this load/ambient). Operator notes "monitor" but does not escalate.
2025-05-25 21:07	Tank-vibration relay action at 4.1 mm/s. Forced outage declared.
2025-05-25 22:30	CPS dispatched. ETA 5 hours due to weekend staffing. EPA-2024-11 invoked 23:15.
2025-05-26 03:10	Apex onsite. Initial inspection finds two loosened core-clamp bolts on the C-phase limb and Buchholz relay residue indicating mild gas evolution.
2025-05-26 — 2025-05-30	Core-clamp re-torque, gasket replacement, Buchholz reset, vibration baseline. Apex recommends transformer replacement; internal engineering counter-proposes field repair.
2025-05-31 — 2025-06-01	Re-energization sequence, 24-hour surveillance, return to service.

3. Telemetry Observations

- Tank vibration at trip: 4.1 mm/s (MS-2025-03 limit 3.5; Siemens Energy Manual SET-2200 limit 5.0).
- Top-oil at trip: 78.2 °C.
- DGA snapshot: acetylene 0.4 ppm (low), ethylene 22 ppm (elevated baseline 8), methane 11 ppm. Pattern consistent with mechanical loosening rather than electrical fault.
- Core-clamp torque on C-phase: measured 142 N·m (spec 240 N·m). Both clamps loosened.
- Quarterly core-clamp inspection (WO-2025-03059) was on the Q1 deferred list. Last documented inspection: 2024-09-12.
- Ambient: 18–22 °C, SW wind 4–8 m/s. Conditions normal.

Vibration baseline post-repair recorded 2.4 mm/s, slightly elevated from commissioning (2.0). The 12% baseline increase persisted to the December event (OIR-2025-004).

4. Restoration Activities

Vendor	Scope	Invoice (USD)
Apex Emergency Grid Solutions	Onsite diagnostics, project management	\$98,000
Apex / Siemens Energy subcontract (1.8x)	Core-clamp re-torque, gasket replacement, Buchholz reset	\$214,000
Total		\$312,000

CPS scheduled equivalent estimate (post-event benchmark): \$82,500. See EMR-2025-003 for the Period A invoice consolidation.

5. Preliminary Root Cause

Direct cause: progressive loosening of two C-phase core-clamp bolts driving sustained tank vibration above the MS-2025-03 alert threshold, ultimately tripping the vibration relay.

Contributing causes:

1. Quarterly core-clamp inspection deferred under the Q1 2025 budget freeze. The last documented torque check was September 2024 — 8 months prior to event.
2. Technician fatigue during the cascade. The lead technician (R. Vega) had logged 67.5 hours that week, exceeding FMP-2 limit of 60 hr/wk. This is documented in CAR-2025-001 (FMP-2 violations during Spring Vibration Cascade).
3. Apex's replacement recommendation appears to follow the pattern observed in other Period A events: emergency vendor consistently recommends replacement over field repair. Internal engineering successfully field-repaired. See EMR-2025-003 and RCA-2025-001.

6. Recommendations

1. Execute fleet-wide core-clamp torque audit on all NW-PowerPool transformers within 60 days. Track under CAR-2025-008.
2. Reinforce FMP-2 fatigue caps during emergency events. Refer to CAR-2025-001.
3. Add Buchholz residue trace evidence to standard post-outage telemetry capture. CSM-7 amendment to be drafted.
4. Reject Apex's transformer replacement quote (\$1.86M). Internal field repair successful; replacement not warranted.

5. Cross-reference OIR-2025-004 (December 2025 overheating event on the same TRA-10561) and RCA-2025-001 (Spring Vibration Cascade root analysis).

Signatures

Role	Name	Date
Author	Elena Vasquez, Reliability Engineer II	2025-06-10
Plant Manager	Priya Ramachandran	2025-06-11
Compliance Review	D. Whitcombe	2025-06-12

Distribution: NW-PowerPool Operations · Reliability Engineering · Compliance · OEM Liaison · Vendor Management · CFO Office

References: OIR-2025-004 · OIR-2025-006 · CAR-2025-001 · CAR-2025-008 · EMR-2025-003 · RCA-2025-001 · Siemens Energy Manual SET-2200 · Standard MS-2025-03 · CSM-7 · FMP-2